



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to K.L. AGENCY

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

#### Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

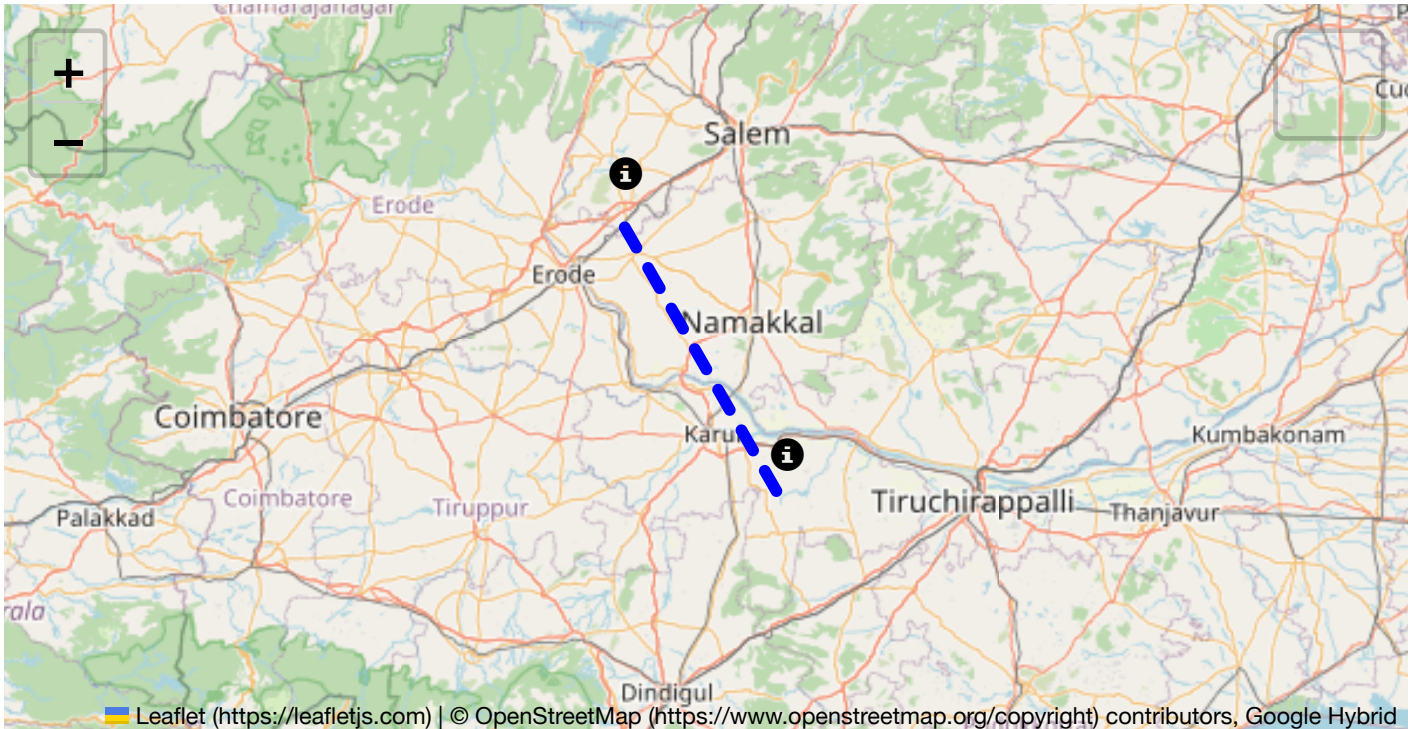
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

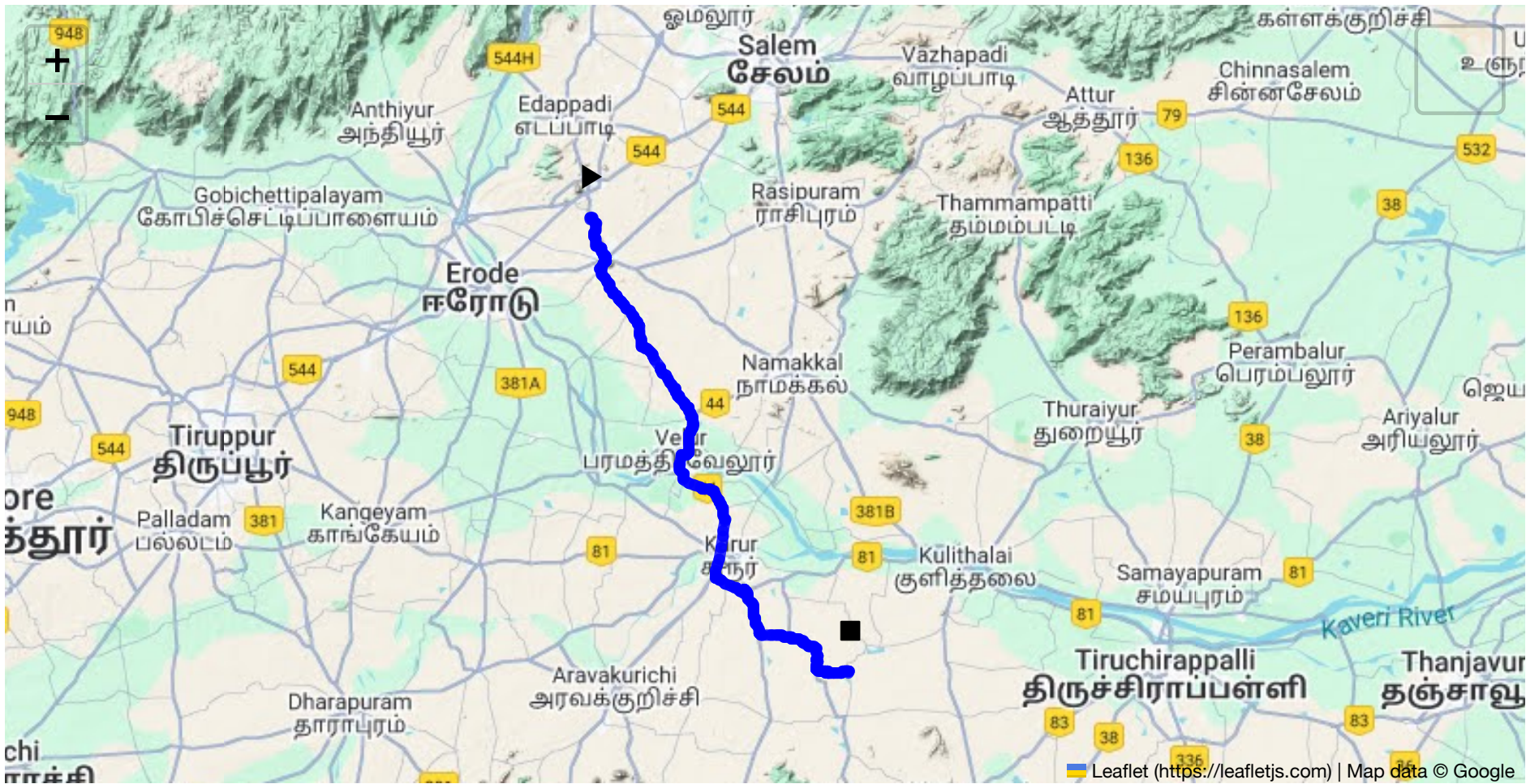
This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 101.67 km**  
**Estimated Duration: 2.1 hours**  
**Adjusted Duration (Heavy Vehicle): 2.7 hours**



Start: (11.4381, 77.8734)  
End: (10.792285, 78.248322)



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from CVQF+23W, Sangagiri, Tamil Nadu to Q6VX+6FX, P.Udayapatti, Pannapatti, Tamil Nadu, covers approximately 101.67 kilometers. The journey typically follows state highways and local roads and winds through both rural and semi-urban areas, consisting mainly of Sathyamangalam Road towards Erode and NH 544 leading towards Namakkal and Pannapatti.

### 2. Typical Weather Conditions and Potential Weather-Related Hazards

The region generally experiences tropical weather with hot summers, moderate monsoons, and mild winters. During the monsoon season (June to September), heavy rainfall may lead to water-logging and reduced visibility on certain road segments, increasing the risk of hydroplaning and accidents. In summer, high temperatures can lead to road surface damage and glare, posing additional challenges.

### 3. Analysis of Traffic Patterns

Traffic patterns along this route vary:

- Peak Hours:** Morning (8 AM to 10 AM) and evening (5 PM to 8 PM) typically see increased traffic, particularly near towns and intersections.
- Congestion-Prone Areas:** Sangagiri and Namakkal are significant congestion points, especially near market areas and junctions, where local traffic mingles with through traffic.

### 4. Assessment of Road Quality and Infrastructure

- Road Quality:** Generally good along NH 544, with some sections of the state highway in need of maintenance. Expect occasional potholes and narrow lanes in rural roads.

- **Infrastructure:** Limited street lighting in rural areas and irregular signage could pose navigation challenges at night.

## 5. Suggestions for Alternative Routes for Emergencies

In emergencies, alternative routes via NH 544 could be considered, bypassing smaller local roads. GPS should be consulted for real-time updates or diversions as conditions warrant.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu follows strict regulations for transporting hazardous materials, including a requirement for placards, proper documentation, and adherence to prescribed routes and times. Night travel is generally discouraged.

## 7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Historical data suggests isolated incidents of heavy vehicle collisions on NH 544, primarily due to over-speeding and mechanical failure. No significant hazardous material spills have been reported recently.

## 8. Environmental Considerations and Sensitive Areas

Key environmental concerns include passing through agricultural lands where crop and soil damage could occur in the case of a spill. Some areas near Namakkal are recognized for their biodiversity; care should be taken to minimize ecological disruption.

## 9. Analysis of Communication Coverage

Mobile network coverage is generally good along NH 544 but diminishes in rural and less populated areas. Potential dead zones may exist in the more remote segments between townships.

## 10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas (Sangagiri, Namakkal):** Emergency services can typically respond within 30 minutes.
- **Rural Segments:** Response times may increase to 45-60 minutes due to distance and road conditions.

## 11. Overall Summary of Risk Assessment

The route poses moderate risks primarily due to weather conditions and traffic congestion. Road quality is generally good, although rural road sections require caution. Communication and timely emergency assistance are reliable in urban areas but may lag in rural segments. To mitigate risks, careful journey planning, adherence to regulations, and situational awareness are essential.

|    | Risk Type  | Risk Level | Coordinates                   | Speed Limit |
|----|------------|------------|-------------------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480            | 15 KM/Hr    |
| 1  | Roundabout | High       | 10.92958, 78.05452            | 15 KM/Hr    |
| 2  | U-Turn     | High       | 10.9110477, 78.09713769999999 | 10 KM/Hr    |
| 3  | Turn       | High       | 11.43968, 77.87345            | 15 KM/Hr    |
| 4  | Turn       | High       | 11.44019, 77.87547            | 15 KM/Hr    |
| 5  | Turn       | Medium     | 11.39685, 77.88861            | 30 KM/Hr    |
| 6  | Turn       | High       | 11.37862, 77.89488            | 15 KM/Hr    |
| 7  | Turn       | High       | 11.37850, 77.89453            | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.16755, 78.01568            | 30 KM/Hr    |
| 9  | Turn       | High       | 11.14545, 78.02208            | 15 KM/Hr    |
| 10 | Turn       | High       | 11.14485, 78.02282            | 15 KM/Hr    |
| 11 | Turn       | Medium     | 11.10084, 78.00555            | 30 KM/Hr    |
| 12 | Turn       | High       | 10.90793, 78.09916            | 15 KM/Hr    |
| 13 | Blind Spot | Blind Spot | 10.91105, 78.09714            | 10 KM/Hr    |
| 14 | Turn       | High       | 10.91103, 78.09733            | 15 KM/Hr    |
| 15 | Turn       | High       | 10.84613, 78.12000            | 15 KM/Hr    |
| 16 | Turn       | Medium     | 10.84321, 78.16436            | 30 KM/Hr    |
| 17 | Turn       | High       | 10.84057, 78.16518            | 15 KM/Hr    |
| 18 | Turn       | High       | 10.83953, 78.18108            | 15 KM/Hr    |
| 19 | Turn       | Medium     | 10.83523, 78.18173            | 30 KM/Hr    |
| 20 | Turn       | Medium     | 10.82892, 78.20166            | 30 KM/Hr    |
| 21 | Turn       | Medium     | 10.82865, 78.20188            | 30 KM/Hr    |
| 22 | Turn       | Medium     | 10.80081, 78.20092            | 30 KM/Hr    |
| 23 | Turn       | High       | 10.79731, 78.20146            | 15 KM/Hr    |
| 24 | Turn       | Medium     | 10.79717, 78.20346            | 30 KM/Hr    |
| 25 | Turn       | Medium     | 10.79674, 78.20768            | 30 KM/Hr    |
| 26 | Turn       | High       | 10.79660, 78.20790            | 15 KM/Hr    |
| 27 | Turn       | High       | 10.79591, 78.20791            | 15 KM/Hr    |
| 28 | Turn       | Medium     | 10.79526, 78.20925            | 30 KM/Hr    |
| 29 | Turn       | High       | 10.79528, 78.21015            | 15 KM/Hr    |
| 30 | Turn       | High       | 10.79561, 78.21019            | 15 KM/Hr    |
| 31 | Blind Spot | Blind Spot | 10.79620, 78.21425            | 10 KM/Hr    |
| 32 | Blind Spot | Blind Spot | 10.79408, 78.21442            | 10 KM/Hr    |
| 33 | Turn       | Medium     | 10.79456, 78.21619            | 30 KM/Hr    |
| 34 | Turn       | High       | 10.79415, 78.21772            | 15 KM/Hr    |
| 35 | Turn       | High       | 10.79248, 78.21772            | 15 KM/Hr    |
| 36 | Turn       | Medium     | 10.79219, 78.23677            | 30 KM/Hr    |
| 37 | Turn       | Medium     | 10.79289, 78.23755            | 30 KM/Hr    |



|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 38 | Turn      | High       | 10.79450, 78.23820 | 15 KM/Hr    |
| 39 | Turn      | Medium     | 10.79425, 78.24209 | 30 KM/Hr    |
| 40 | Turn      | Medium     | 10.79517, 78.24493 | 30 KM/Hr    |
| 41 | Turn      | Medium     | 10.79556, 78.24534 | 30 KM/Hr    |
| 42 | Turn      | High       | 10.79561, 78.24564 | 15 KM/Hr    |
| 43 | Turn      | High       | 10.79512, 78.24573 | 15 KM/Hr    |
| 44 | Turn      | Medium     | 10.79490, 78.24612 | 30 KM/Hr    |
| 45 | Turn      | High       | 10.79500, 78.24761 | 15 KM/Hr    |

### Emergency Locations

|    | type     | name                             | coordinates            | speed_limit | risk_level |
|----|----------|----------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885, 77.8965774 | 30 km/h     | Medium     |
| 6  | clinic   | Kongu Nursing Home               | 11.3783065, 77.8961134 | 30 km/h     | Medium     |
| 7  | hospital | Soorya Multispecialty Hospital   | 11.3786429, 77.8931912 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426     | 30 km/h     | Medium     |
| 9  | hospital | Krishna Hospital, Namakkal       | 11.3754811, 77.8931817 | 30 km/h     | Medium     |
| 10 | hospital | Tirukumaran Hospitals            | 11.3782118, 77.8914933 | 30 km/h     | Medium     |
| 13 | police   | கரட்டுப்பாளையம் காவல் நிலையம்    | 11.357586, 77.8922539  | 30 km/h     | Medium     |
| 14 | hospital | Goverment Hospital, Nallur       | 11.2679698, 77.946863  | 30 km/h     | Medium     |
| 16 | hospital | Goverment Hospital               | 11.156647, 78.0213586  | 30 km/h     | Medium     |
| 18 | hospital | Government Hospital, Manmangalam | 11.0275504, 78.0646437 | 30 km/h     | Medium     |

### Crowded Spots

|   | type        | name                                | coordinates            | speed_limit | risk_level |
|---|-------------|-------------------------------------|------------------------|-------------|------------|
| 2 | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி        | 11.3850439, 77.8948279 | 30 km/h     | Medium     |
| 3 | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி       | 11.3844247, 77.8949053 | 30 km/h     | Medium     |
| 4 | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை | 11.3833608, 77.8970145 | 30 km/h     | Medium     |

|    | type    | name                                                           | coordinates               | speed_limit | risk_level |
|----|---------|----------------------------------------------------------------|---------------------------|-------------|------------|
| 11 | school  | KSR Educational institution                                    | 11.3772013,<br>77.8908807 | 30 km/h     | Medium     |
| 12 | school  | MDV School                                                     | 11.3719843,<br>77.8915244 | 30 km/h     | Medium     |
| 15 | school  | Sivabakkiam Muthusamy Matriculation<br>Higher Secondary School | 11.1788177,<br>78.0061696 | 30 km/h     | Medium     |
| 17 | school  | Goverment Higher Secondary School                              | 11.1511724,<br>78.0200723 | 30 km/h     | Medium     |
| 19 | college | valluvar catering                                              | 10.9553473,<br>78.0606583 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.38395, 77.89480



Risk Type: Roundabout  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 10.92958, 78.05452





**Risk Type:** U-Turn

**Risk Level:** High

**Speed Limit:** 10 KM/Hr

**Coordinates:** 10.9110477, 78.09713769999999



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44019, 77.87547



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.39685, 77.88861





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.37862, 77.89488



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.37850, 77.89453



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.16755, 78.01568





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.14545, 78.02208



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.14485, 78.02282



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.10084, 78.00555





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.90793, 78.09916



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 10.91105, 78.09714



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.91103, 78.09733





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.84613, 78.12000



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.84321, 78.16436



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.84057, 78.16518





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.83953, 78.18108



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.83523, 78.18173



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.82892, 78.20166





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.82865, 78.20188



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.80081, 78.20092



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79731, 78.20146





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79717, 78.20346



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79674, 78.20768



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79660, 78.20790





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79591, 78.20791



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79526, 78.20925



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79528, 78.21015





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79561, 78.21019



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 10.79620, 78.21425



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 10.79408, 78.21442





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79456, 78.21619



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79415, 78.21772



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79248, 78.21772





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79219, 78.23677



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79289, 78.23755



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79450, 78.23820





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79425, 78.24209



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79517, 78.24493



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79556, 78.24534





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79561, 78.24564



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79512, 78.24573



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.79490, 78.24612





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.79500, 78.24761

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